

UPI FRAUD STATISTICAL ANALYSIS

Statistical Signals for Detecting Fraudulent UPI Transactions

PROJECT REPORT

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ABSTRACT

UPI (Unified Payments Interface) has become one of the most widely used digital payment methods. With the increasing number of digital transactions, detecting fraudulent transactions has become important. This project focuses on analysing UPI transaction data using statistical methods and data visualization techniques. The dataset contains transaction-related features such as transaction amount, transaction hour, failed attempts, new device usage, unusual location, transaction frequency, and fraud status. Statistical analysis and visualizations are used to identify patterns and signals associated with fraudulent transactions. The project demonstrates how data analysis can help understand suspicious transaction behaviour and support the development of fraud detection systems.

INTRODUCTION

UPI (Unified Payments Interface) is a digital payment system that allows users to transfer money quickly and easily using mobile devices. The rapid growth of UPI has increased the number of digital transactions and made secure transaction monitoring more important. Fraudulent transactions can cause financial loss and reduce user trust in digital payment systems.

This project analyses UPI transaction data using statistical analysis and data visualization techniques. The study examines transaction amount, transaction hour, failed attempts, new device usage, unusual location, transaction frequency, and fraud status. By analysing these factors, the project aims to identify statistical signals that may be associated with fraudulent transactions.

The results of this analysis can help in understanding suspicious transaction patterns and can support the development of effective fraud detection systems.

PROBLEM STATEMENT

The increasing use of UPI and digital payment services has also increased the need to identify suspicious transactions. Fraudulent transactions may have different patterns related to transaction amount, failed attempts, new devices, unusual locations, and frequent transactions. Therefore, analysing transaction data using statistical methods is important for identifying signals that may indicate fraudulent activity. This project aims to study these patterns and understand how statistical analysis can support UPI fraud detection.

OBJECTIVES

1. To analyse UPI transaction data using statistical methods.
2. To identify patterns associated with fraudulent transactions.
3. To study the relationship between transaction features and fraud.
4. To analyse factors such as failed attempts, new devices, unusual locations, and transaction frequency.
5. To use data visualization techniques to understand fraud patterns.
6. To identify statistical signals that can support UPI fraud detection.

DATASET DESCRIPTION

The dataset used in this project contains 1,000 UPI transactions. Each transaction contains information related to transaction behaviour and fraud status. The main features include transaction ID, transaction amount, transaction hour, failed attempts, new device usage, unusual location, transaction count within one hour, and fraud status.

The fraud column is the target variable, where 0 represents a normal transaction and 1 represents a fraudulent transaction. These features are analysed to identify patterns and statistical signals related to UPI fraud.

TOOLS AND TECHNOLOGIES

1. Python – Used for data analysis and statistical calculations.
2. Pandas – Used for loading, cleaning, and analysing the dataset.
3. NumPy – Used for numerical operations.
4. Matplotlib – Used to create charts and graphs.
5. Seaborn – Used to create statistical visualizations.
6. Jupyter Notebook – Used to write and execute Python analysis code.
7. Google Docs – Used to prepare the project report.

METHODOLOGY

The methodology of this project consists of several steps. First, the UPI transaction dataset is loaded into a Python environment using Pandas. The dataset is then examined to understand its structure and important features.

Next, statistical analysis is performed to calculate fraud counts, fraud percentage, and basic statistical measures. Different transaction features are compared with the fraud status to identify possible fraud-related patterns.

Data visualization techniques such as count plots and correlation heatmaps are used to present the results clearly. Finally, the findings are analysed to identify statistical signals that may help in understanding UPI fraudulent transactions.

STATISTICAL ANALYSIS

The statistical analysis was performed to understand the distribution of fraudulent and normal UPI transactions.

1. Fraud Transaction Count

The dataset contains 811 fraudulent transactions and 189 normal transactions.

2. Fraud Percentage

The percentage of fraudulent transactions in the dataset is 81.1%.

3. Basic Statistical Analysis

Descriptive statistics were used to understand the numerical features of the dataset, including transaction amount, transaction hour, failed attempts, and transaction frequency.

4. Fraud Factor Analysis

The relationship between fraud and different transaction features was analysed using cross-tabulation. The factors studied include failed attempts, new device usage, unusual location, transaction frequency, and transaction hour.

5. Correlation Analysis

Correlation analysis was performed to understand the relationship between numerical transaction features and the fraud variable.

DATA VISUALIZATIONS

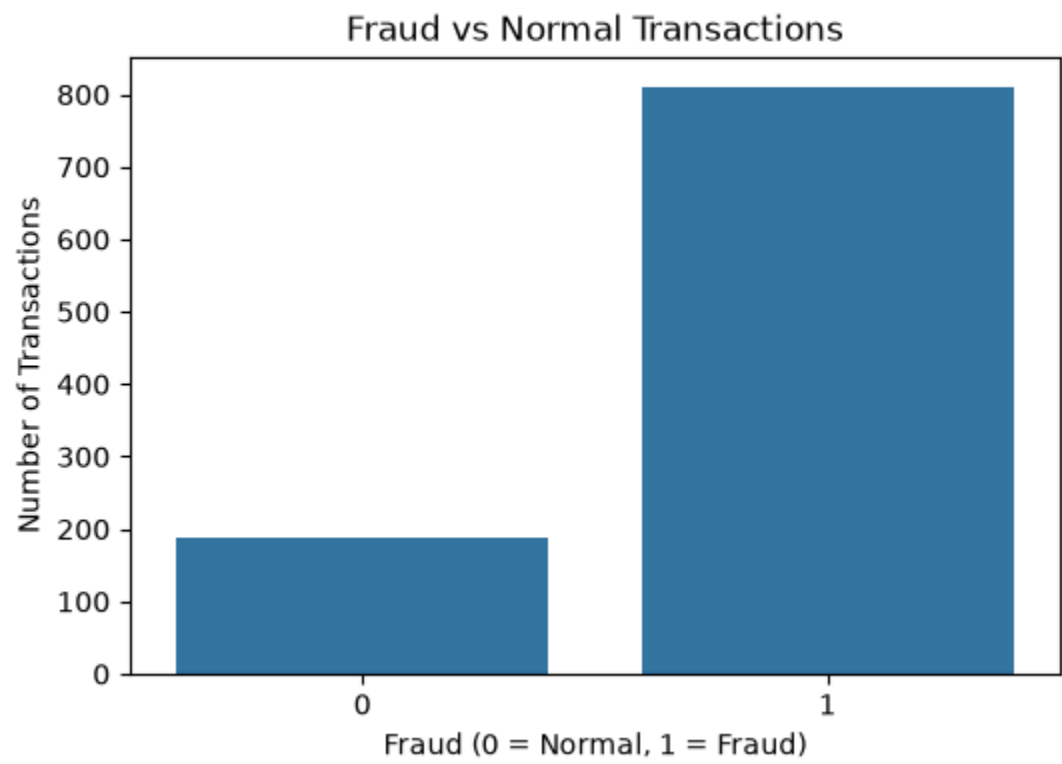


Figure 1: Fraud vs Normal Transactions

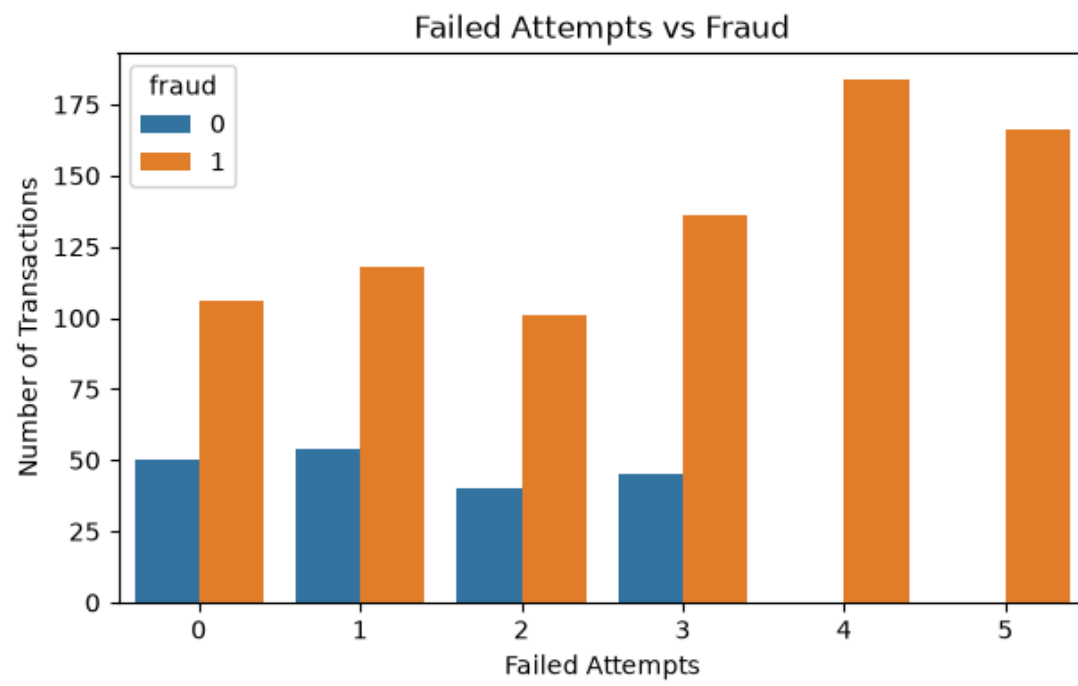


Figure 2: Failed Attempts vs Fraud

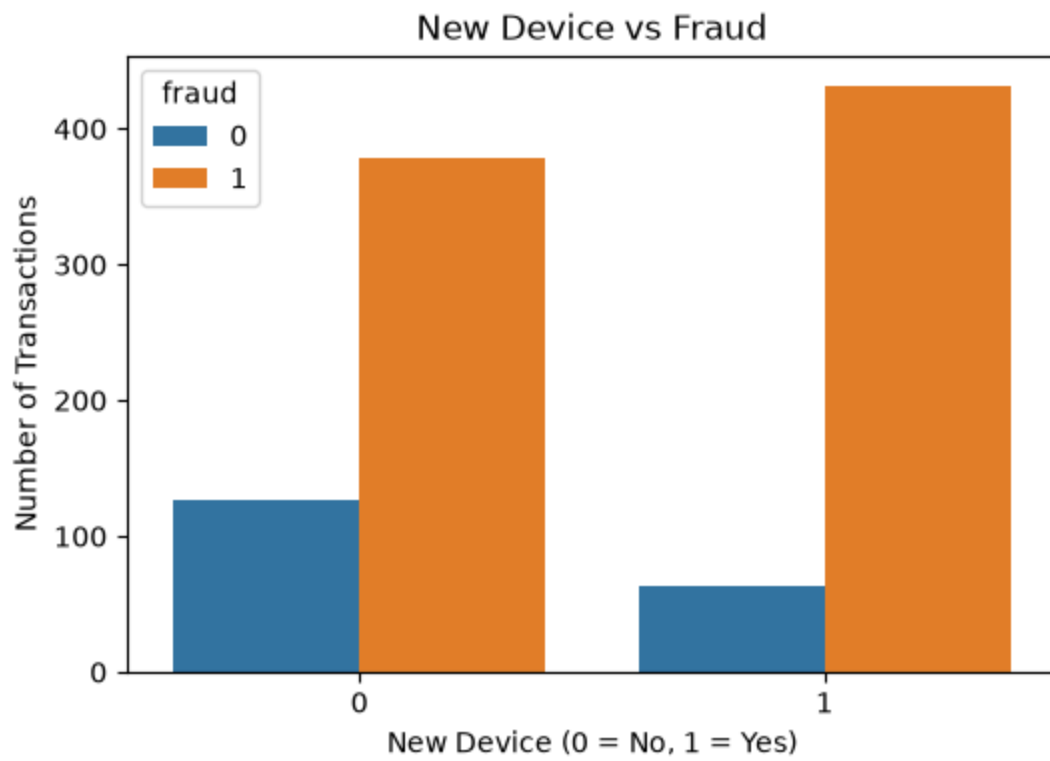


Figure 3: New Device vs Fraud

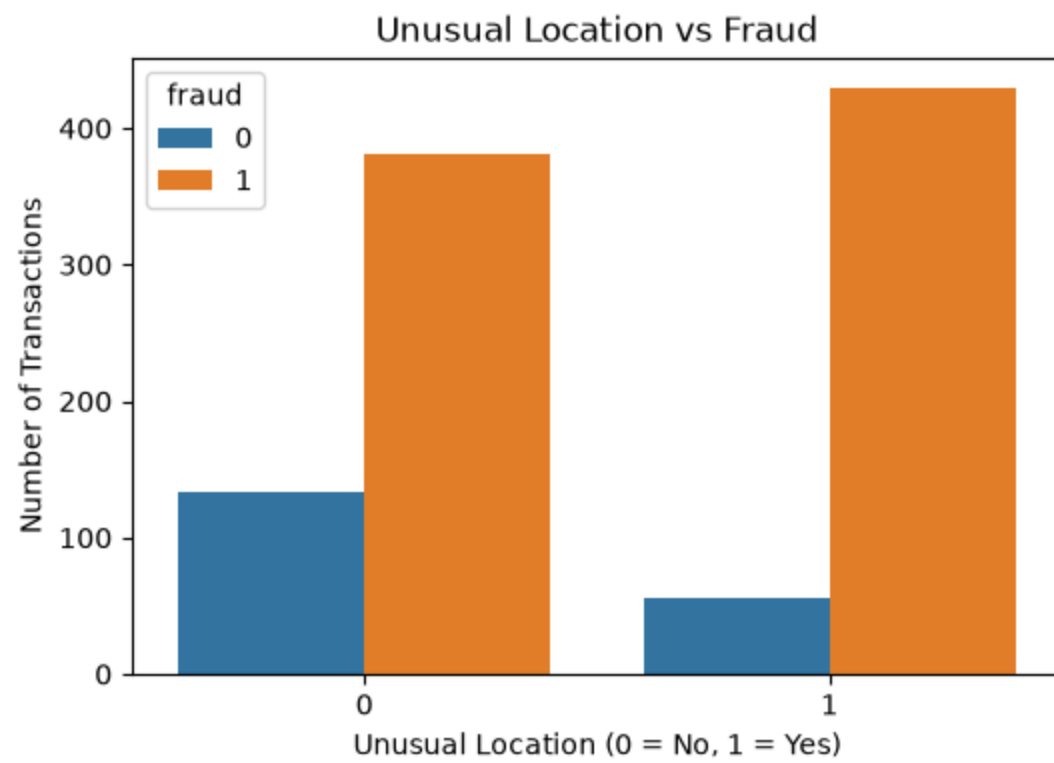


Figure 4: Unusual Location vs Fraud

DATA VISUALIZATIONS

The analysis of 1,000 UPI transactions provided useful insights into fraudulent transaction patterns. Among the transactions analysed, 811 were identified as fraudulent and 189 were normal transactions. This indicates that 81.1% of the transactions in the dataset were classified as fraud.

The analysis also examined different transaction features such as failed attempts, new device usage, unusual location, transaction frequency, and transaction hour. These features were compared with the fraud status using statistical analysis and visualizations.

The results show that transaction behaviour can provide useful signals for identifying suspicious transactions. Statistical analysis and graphical representations help in understanding these patterns and can support the development of fraud detection systems.

CONCLUSION

This project analysed UPI transaction data using statistical analysis and data visualization techniques. The study examined different transaction features such as failed attempts, new device usage, unusual location, transaction frequency, transaction hour, and transaction amount.

The analysis identified useful statistical signals associated with fraudulent transactions. The results show that transaction behaviour can provide important information for detecting suspicious activities. Data visualization also helped in understanding the patterns more clearly.

Overall, statistical analysis can be a useful approach for studying UPI fraud patterns and supporting the development of effective fraud detection systems.

FUTURE ENHANCEMENT

1. The project can be enhanced by using larger and real-world UPI transaction datasets.
2. Machine learning algorithms can be added to automatically classify transactions as normal or fraudulent.
3. A real-time fraud detection system can be developed to identify suspicious transactions immediately.
4. More transaction features can be included to improve fraud detection accuracy.
5. A dashboard can be developed to display fraud statistics and suspicious transaction patterns interactively.

REFERENCES

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2. Pandas Documentation – Data analysis and data manipulation.
3. NumPy Documentation – Numerical computing with Python.
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5. Seaborn Documentation – Statistical data visualization.
6. Jupyter Documentation – Interactive notebook environment.